



FINAL REPORT - PRODUCT RELEASED

Boston, MA, United States

Origin



Frankfurt, Germany

Destination

1. EVENT IDENTIFICATION

INVESTIGATION EXC-VC-1026-04	TEMPERATURE RECORD TL-VCL-1026	SHIPMENT SHIP-VC-1026-BOS-FRA	POD VCL-1026-8841
CUSTOMER Caldova Therapeutics	LANE AUTHORIZATION LA-VCL-1026	WAYBILL 020-8841-1026	SERVICE Validated 2-8 C air shipment

2. EXCURSION DETAILS

EVENT TYPE High temperature	STARTED 15 Oct 2026 08:30 UTC	RETURNED IN RANGE 15 Oct 2026 11:00 UTC	DURATION 2 hr 30 min
VALIDATED RANGE 2.0-8.0 C	MAXIMUM RECORDED 10.7 C	LOCATION Frankfurt airport controlled-transfer area	

Event description: Temperature exceeded the validated upper limit during the ground-handler transfer from the aircraft container to refrigerated road transport.

3. AFFECTED INVENTORY AND IMMEDIATE ACTION

PRODUCT Caldova temperature-controlled clinical supply	LOT CLD-VCL-1026-A	QUANTITY AFFECTED 12 insulated cartons (4,800 units)
PREVIOUS EXCURSIONS None documented	PHYSICAL QUARANTINE Yes - confirmed	DETECTED 15 Oct 2026 14:12 UTC

- ☑ Shipment moved to qualified 2-8 C storage at 11:06 UTC.
- ☑ All affected shipping units labeled QUARANTINE - DO NOT DISTRIBUTE.
- ☑ Caldova quality contact notified at 14:25 UTC.
- ☑ Logger file and chain-of-custody records secured for assessment.

4. EVIDENCE REVIEWED

EVIDENCE	IDENTIFIER	FINDING
USB logger export	TL-VCL-1026	One high event; peak 10.7 C; 2 hr 30 min above limit.
Proof of delivery	VCL-1026-8841	Shipment received intact; no visible shipper damage recorded.
Lane authorization	LA-VCL-1026	Boston-Frankfurt 2-8 C lane approved for October operation.



5. INVESTIGATION

Probable cause

The active air container remained at the controlled-transfer bay for 94 minutes beyond the validated handoff window after a ground-handler vehicle change.

Contributing factor

The replacement refrigerated vehicle arrived after the scheduled connection window.

CAUSE CATEGORY

Ground-handler transfer delay

PACKAGING CONDITION

Intact; no opening or physical damage observed

6. PRODUCT QUALITY ASSESSMENT

ASSESSMENT REQUESTED

15 Oct 2026 15:05 UTC

ASSESSMENT COMPLETED

16 Oct 2026 17:40 UTC

STABILITY CASE

CAL-QA-2026-184

CUMULATIVE EXPOSURE

Reviewed - no prior events

ASSESSMENT CONCLUSION

The recorded exposure remained within the approved stability allowance for lot CLD-VCL-1026-A.

7. CORRECTIVE AND PREVENTIVE ACTIONS

ITEM	ACTION	OWNER	DUE
CA-1	Require a confirmed backup vehicle before aircraft-container release.	EU Lane Operations	30 Oct 2026
CA-2	Escalate transfer delays at 30 minutes rather than 60 minutes.	EU Lane Operations	30 Oct 2026
CA-3	Add ground-handler handoff time to the monthly lane performance review.	Quality Systems	15 Nov 2026

8. FINAL DISPOSITION

☒ Released for distribution

The recorded exposure remained within the approved stability allowance for lot CLD-VCL-1026-A. Quarantine was removed on 16 Oct 2026 18:10 UTC. Product may proceed through the authorized distribution channel.

9. APPROVALS

Jonas Keller, Valence Quality Operations

Investigator | Signed 16 Oct 2026

Priya Desai, Valence Responsible Person

Valence quality review | Signed 16 Oct 2026

Mara Levin, Caldova Director of Product Quality

Customer disposition approval | Signed 16 Oct 2026

10. ATTACHED AND RELATED RECORDS

- Temperature logger report TL-VCL-1026
- Proof of delivery VCL-1026-8841
- Lane authorization LA-VCL-1026
- Supplier invoice INV-SUP-008-2026-10, excursion handling line L003